

# AI Chat Assistant Report

## Introduction

This project is an AI Chat Assistant developed using Python, Streamlit, LangChain, and Groq API. The application provides an interactive chatbot interface where users can communicate with an AI model in real time. The system is designed with a modern dark-themed interface and maintains conversation history during the session.

## Project Objective

The main objective of this project is to create a smart conversational AI assistant capable of answering user questions clearly and efficiently using Large Language Models (LLMs).

## Technologies Used

- Python
- Streamlit
- LangChain
- Groq API
- Llama 3.3 70B Versatile Model
- HTML/CSS

## System Workflow

- The user enters a message through the Streamlit interface.
- The message is sent to the LangChain prompt template.
- Previous chat history is included in the conversation context.
- The Groq LLM processes the request.
- The AI response is returned and displayed on the screen.
- The conversation is stored using Streamlit Session State.

## Code Explanation

### Environment Variables

The project uses a .env file to securely store the Groq API key.

```
load_dotenv(override=True)
```

```
GROQ_API_KEY = os.getenv("GROQ_API_KEY")
```

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### **Prompt Template**

The system prompt defines the AI assistant behavior and personality.

```
template = ChatPromptTemplate.from_messages(...)
```

The prompt also includes:

- System instructions
  - Chat history
  - User input
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### **Language Model Integration**

The project integrates the Groq API using LangChain.

```
llm = ChatGroq(  
    model="llama-3.3-70b-versatile",  
    GROQ_API_KEY=GROQ_API_KEY  
)
```

The selected model is optimized for fast and accurate responses.

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### **Chain Architecture**

The processing pipeline combines:

- Prompt Template
- Language Model
- Output Parser

```
chain = template | llm | parser
```

This architecture simplifies the flow of user requests and AI responses.

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### **User Interface**

The frontend is built using Streamlit with custom CSS styling. The interface uses:

- Dark background colors
- Styled message containers
- Separate user and AI message designs

The application also supports:

- Real-time chatting
  - Conversation clearing
  - Dynamic message updates
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### **Chat History Management**

The application uses `st.session_state` to preserve chat history during the session.

`st.session_state.chat_history`

`st.session_state.display_history`

This allows the AI assistant to remember previous messages and maintain conversational context.

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### **Advantages of the Project**

- Simple and user-friendly interface
  - Fast AI responses using Groq
  - Conversation memory support
  - Professional UI design
  - Easy to customize and expand
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### **Future Improvements**

Possible future enhancements include:

- Voice input support
  - Database integration
  - Streaming AI responses
  - Authentication system
  - Multi-model selection
  - File upload and RAG integration
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### **Conclusion**

This project demonstrates the integration of modern AI technologies to build an intelligent chatbot application. By combining Streamlit, LangChain, and Groq API, the

system delivers an efficient and responsive conversational experience with an attractive user interface.